BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Slope of a line ax + by + c = 0 is:

(A) -a/b
(B) a/b
(C) -b/a
(D) b/a

Q2. Inclination of a line x = 6 is:

(A) 0
(B) π
(C) π/4
(D) π/2

Q3. The point of intersection of lines y = 2 and x = -1 is:

(A) (2,-1)
(B) (2,1)
(C) (2,0)
(D) (-1,2)

Q4. x - y < 2 is satisfied by the point:

(A) (3,1)
(B) (-1,1)
(C) (1,-1)
(D) (0,-2)

Q5. Center of a circle $(x+1)^2 + y^2 = 25$ is:

(A) (0,0)
(B) (1,0)
(C) (-1,0)
(D) (0,2)

Q6. Major axis of $x^2/a^2 + y^2/b^2 = 1$

(A) x = a
(B) x = ±[illegible in source scan]
(C) y = ±a/b
(D) y = 0

Q7. Length of minor axis of $x^2 + 4y^2 = 16$ is:

(A) 4 (B) 16

(C) 20 (D) 16

Q8. Center of $(x-1)^2/4 - (y+1)^2/16 = 1$ is:

(A) (0,0) (B) (1,-1)

(C) (-1,-1) (D) (-1,1)

Q9. The direction cosines of y-axis are:

(A) 0,0,0 (B) 1,0,0

(C) 0,1,0 (D) 0,0,1

Q10. $i.k \times j = ?$

(A) 0 (B) 1

(C) 1/2 (D) -1

Q11. $\lim(x\to0)\ x/\sin2x = ?$

(A) 1/2 (B) 2

(C) -1/2 (D) -2

Q12. The function $f(x) = x/(x^2 - 4)$ is discontinuous at:

(A) 0 (B) ±2

(C) 1 (D) ±1

Q13. $\lim(h\to0)\ (f(a+h) - f(a))/h = ?$

(A) f(a) (B) f'(a+h)

(C) f'(a) (D) f'(x)

Q14. $dy/dx = -1/x^2$ if:

(A) $y = -1/x$ (B) $y = -x$

(C) $y = \ln x$ (D) $y = 1/x$

Q15. If $f(x) = \cos x$, then $f'(\pi/2) = ?$

(A) -1 (B) 1

(C) 0 (D) 2

Q16. $d/dx(e^{(1/x)}) = ?$

(A) $e^{(1/x)}$ (B) $-1/x^2\ e^{(1/x)}$

(C) $1/x^2\ e^{(1/x)}$ (D) $1/x^2\ e^{(x^{(-1/2)})}$

Q17. Differential of √y is:

(A) √y dx
(B) √y dy
(C) 1/(2√y) dy
(D) 1/(2√y) dx

Q18. ∫sec ax tan ax dx = ? is:

(A) tan ax/a + c
(B) sec ax/a + c
(C) sec ax tan ax/a + c
(D) cot ax/a + c

Q19. $\int 1/(x^2 + 3)\, dx$ = ?

(A) $\tan^{-1}(x/3) + c$
(B) $1/3 \tan^{-1}(x/3) + c$
(C) $1/\sqrt{3} \tan^{-1}(x/\sqrt{3}) + c$
(D) $\cot^{-1}(3x) + c$

Q20. ∫ from -π/2 to π/2 cos(x/2) dx = ?

(A) 1
(B) √2
(C) 2√2
(D) 4√2

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80

Note: Note: Section I is compulsory. Attempt any three (3) questions from section II.

Note: SECTION-I

Q2.

(i) Define constant function with example.

(ii) Find $f^{-1}(x)$ if f(x) = -2x + 8

(iii) Evaluate $\lim_{x\to 1} (x^4 - 3x^2 + 3x - 1)/(x^3 - x)$

(iv) Find derivative by definition $1/x^{40}$.

(v) Differentiate w.r.t. x, √(1+x)/√(1-x)

(vi) Find dy/dx, $xy + y^2 = 2$

(vii) Differentiate w.r.t. x, cos√x + √sin x

(viii) Differentiate $\cos^{-1}(x/a)$

(ix) Find dy/dx if y = xe^(sin x)

(x) Determine the interval in which f(x) = cos x is increasing or decreasing for the domain x = (-π/2, π/2)

(xi) Define problem constraint.

(xii) Graph the solution set of $5x - 4y \leq 20$

Q3.

(i) Find the area bounded by cos, function from $x = -\pi/2$ to $x = \pi/2$

(ii) Solve the differential equation $(e^x + e^{-x})\, dy/dx = e^x - e^{-x}$

(iii) Evaluate $\int$ from -1 to 2 $(x + |x|)\, dx$

(iv) Evaluate $\int e\hat{}(m \tan^{-1}x)/(1 + x^2)\, dx$

(v) Evaluate $\int 1/(x \ln x)\, dx$

(vi) Evaluate $\int (1 - \sqrt{x})^2/\sqrt{x}\, dx$

(vii) Use differential find dy/dx if $x^2 + 2y^2 = 16$.

(viii) Find the value of $2i \times 2j.k$

(ix) Find a unit vector in the direction of $v \times i + 2j - k$

(x) If AB vector = CD vector, then find the coordinate of the point A when points, B, C, D are (1,2), (-2,5), (4,11) respectively

(i) Find the volume of parallel piped if $u = 3i + 2k$, $v = i + 2j + k$, $w = -j + 4k$.

(ii) Find a vector of length 5, in the direction opposite that of $v = i - 2j + 3k$

Q4.

(i) Find the midpoint of the line segment joining the points A(3,1); B (-2,-4). Also find the distance between them.

(ii) Find slope and inclination of the line joinin the points (3,-2);(2,7)

(iii) Find an equation of horizontal line through (7,-9).

(iv) Find an equation of the line bisecting second and fourth quadrant.

(v) Check whether the lines $3x + 4y - 7 = 0$, $2x - 5y + 8 = 0$, $x + y - 3 = 0$ are concurrent of not?

(vi) Find equation of lines represented by $10x^2 - 23xy - 5y^2 = 0$

(vii) Find the measure of the angle between the lines represented by $3x^2 + 7xy + 2y^2 = 0$

(viii) Find the center and radius of the circle $x^2 + y^2 + 12x - 10y = 0$

(ix) Show that the line $3x - 2y = 0$ is tangent to the circle $x^2 + y^2 + 6x - 4y = 0$

(x) Check the position (5,6) with respect to the circle $2x^2 + 2y^2 + 2x + 8y + 1 = 0$

(xi) Find focus and directrix of the parabola $y^2 = 8x$

(xii) Find an equation of ellipse with given data. Vertices (-1,1), (5,1); foci (4,1) and (0,1)

(xiii) Find the Equation of Hyperbola with given data. Foci (0,±6), e = 2

> Note: SECTION-II

Q5.

(a) Evaluate Lim(x→0) (sec x - cos x)/x

(b) If y = √x - 1/x, then show that 2x dy/dx + y = 2√x

Q6.

(a) Evaluate ∫e^(2x) cos 3x dx

(b) Find an Equation of line through intersection of lines x - y - 4 = 0, 7x + y + 20 = 0 and parallel to line 6x + y - 14 = 0

Q7.

(a) Evaluate: ∫ from 1/2 to √3/2 $\sin^{-1}x/\sqrt{(1-x^2)}$ dx

(b) Maximize f(x,y) = 2x + 5y subject to the constraints: 2y - x ≤ 8; x - y ≤ 4; x ≥ 0; y ≥ 0

Q8.

(a) Show that $y = x^x$ has a minimum value at x = 1/e.

(b) Find an equation of the circle which passes through the points A(5,10), B(6,9) and C (-2,3)

Q9.

(a) Find and equation of the ellipse with given data center (2,2), major axis parallel to y-axis and of length 8 units, minor axis parallel to x-axis and of length 6 units.

(b) Prove that by vector method. sin(α - β) = sin α cos β - cos α sin β